

Muhammad Farrukh

Full Stack Engineer

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Fullstack Engineer with extensive experience in designing and developing scalable, high-performance web applications. Proven ability to optimize system performance, reduce latency, and enhance user experience. Skilled in full-cycle development using JavaScript, TypeScript, ReactJS, NextJS, NodeJS, NestJS, and AWS. Expertise in DevOps, Docker, and CI/CD implementation to streamline deployments and ensure system stability. Adept at delivering high-quality software solutions through agile methodologies and automated testing.

SKILLS

- **Frontend:** ReactJS, NextJS, AngularJS, VueJS, Redux, MUI, Styled Components, RxJS, React Query, Redux Saga, Redux Thunk, Jest, Enzyme, Cypress, Ant Design, TailwindCSS, Typescript, JavaScript, Gatsby, CSS3, HTML5
- **Backend:** NodeJs, NestJS, ExpressJS, Redis, Docker, GraphQL, CI/CD(Jenkins, Github Actions), Mocha, Microservices Architecture, JWT, RabbitMQ, Kafka
- **Databases & Cloud:** AWS (EC2, Lambda, S3,RDS, SQS, SNS), GCP, Heroku, Vercel, MySQL, PostgreSQL, MongoDB, DynamoDB, ORM(Prisma, TypeORM & Sequelize), Apollo GraphQL Server, Payment Integrations (Stripe, Paypal), Git, Sendgrid, 3rd Party API Integration, Socket.IO, GitLab Firebase, WebRTC, SendGrid.

WORK EXPERIENCE

Senior Software Engineer

June 2024 – Present

Terrasmart, California, US

- Configured AWS IoT Core for seamless communication between IoT devices at a Solar Site, optimizing device interaction and data collection.
- Enhanced application performance by 20% through caching (**Redis**) and query optimization techniques.
- Worked with MQTT commands to communicate with the network controller for site management, improving communication reliability and data synchronization between devices.
- Assisted the team in building CI/CD pipelines, streamlining deployments and improving code integration efficiency.
- Utilize Redux Toolkit Query, React Query & Redux Saga, reducing API request failure rates by **15%** and increasing state change efficiency by 22%.
- Implemented user authentication and authorization using **AWS Cognito**, enabling secure sign-up, sign-in, and access control for web and mobile applications.
- Configured multi-factor authentication (MFA), password policies, and user attribute management to enhance application security.
- Utilized **AWS Secrets Manager** to securely store, retrieve, and manage sensitive information such as API keys, database credentials, and configuration secrets.

Senior Software Engineer

June 2024 – Dec 2024

Greeka, Greece, Remote

- Revamped a ferry booking system by migrating from PHP to **NestJS** with **TypeORM** and PostgreSQL.
- Implement RabbitMQ for inter-service communication, enabling seamless message handling across multiple services.
- Designed responsive front-end interfaces with **NextJS** and Tailwind CSS improving user experience.
- Integrated third-party payment gateways, including **Stripe API**, to enable secure and seamless payment.
- Implemented **JWT** authentication and enhanced security protocols, reducing vulnerabilities by 30%.
- Implemented decorators for request validation, authorization checks, and logging to streamline middleware processes.
- Used **TypeScript** to enforce strict typing and catch errors early during development.

Senior Software Engineer

June 2022 – May 2024

Devsinc, San Francisco, US

- Designed and deployed a microservices architecture for a high-traffic e-commerce platform using **NodeJS**.
- Built dynamic and responsive user interfaces using **VueJS & ReactJS**, enhancing user interaction speed by **30%** and reducing bounce rates by 12%
- Integrated AWS S3, Cloudinary for secure, scalable file storage, storing and processing files.
- Designed a multi tenant infrastructure with load balancing and API gateway, scaling the platform to handle concurrent users seamlessly.
- Improved system performance with caching techniques (**Redis**) and query optimization, boosting performance by 20%. Use **JavaScript & TypeScript** to enhance code performance and improve maintainability.
- Integrated MongoDB for database management, handling large-scale data and enabling seamless data storage for E-commerce applications

Software Engineer

Feb 2020 – June 2022

Bitnine Global, Vancouver, CA

- Led the development of Bitnine's official website using **NextJS**, improving page load speed by 30% and enhancing overall user experience.
- Developed and deployed full-stack web applications using **React, Node.js**, and Express.js, improving overall system performance by **20%** through efficient code practices and optimized APIs.
- Spearheaded backend development of RESTful, & **GraphQL** APIs, optimizing data transfer and improving system performance by 20%.
- Created reusable React components, accelerating development speed by 30%.
- Implemented state management using Redux, React Query, and Zustand, improving application maintainability.

PROJECTS

Terrasmart

<https://www.terrasmart.com/>

Developed and maintained front-end features using React.js, delivering responsive, accessible, and user-friendly interfaces. Integrated with AWS IoT Core to enable real-time communication between the web platform and edge devices using MQTT protocol. Designed and deployed serverless functions using AWS Lambda, reducing infrastructure overhead.

Technologies: NodeJs, GraphQL, Typescript, and PostgreSQL, ReactJs, AWS(S3, Lambda, SQS, SNS)

Maple HR

<https://maplehr.io/>

Designed and deployed REST APIs to streamline team operations, reducing administrative overhead by 25%.

Transitioned state management from Redux Thunk to Redux Toolkit Query (RTK Query), reducing boilerplate code.

Technologies: React JS, Node JS, PostgreSQL, Material UI, StoryBook, Typescript, AWS S3

Upmail

<https://www.upmailsolutions.com/>

Led the implementation of a feature that enabled adding multiple dynamic tables within the platform, allowing users to create and manage data sets with flexible inputs and structures, enhancing reporting flexibility and improving data organization.

Technologies: Typescript, React JS, Node JS, Express JS, AWS

EDUCATION

| BACHELOR DEGREE IN COMPUTER SCIENCE